

HRI Multimodal Intent Dataset — Scenario Table V3 (Final)

Adaptive Human–Robot Interaction through Multimodal Emotion, Gesture, and Context-Aware Policy Learning

62 scenarios (31 classroom / 31 kitchen) · 41 train / 21 test (2:1) · 50 videos × 4 actors per scenario · 3,150 clips total · 25 NEW scenarios

1. Dataset design & training protocol

- Split: 42 train / 21 test scenarios (2:1), every intent ≥ 4 train and 2 test rows, both environments covered for all 10 intents.
- Validation: hold out 10 of the 50 videos per TRAIN scenario as the dev set (early stopping / hyperparameters). Never tune on the 21 test scenarios.
- Subjects: 4 actors appear in all splits — report this as a limitation, or (better) reserve one actor's videos for test-only to report a subject-independence score.
- Modality dropout: during training, randomly mask each cue with $p \approx 0.15$ (max 1–2 cues per sample). This teaches the [MISSING] embedding and closes most of the missing-modality gap. When a synthetic mask removes ≥ 2 cues and the remaining cues no longer identify the intent, relabel that augmented sample to the safe default F05 — unless direction still identifies F02/F06 (see rows 58–59).
- none $\neq$ [MISSING]: 'none' is a captured observation (gesture model's idle class); [MISSING] means the cue was not captured at all. The fusion model must treat them differently.
- Fusion models: the table supports all three planned baselines. Gradient-boosted trees: one-hot the 5 categorical inputs (emotion, gesture, motion, direction, context) with 'missing' as a category. Self-attention: 5 cue tokens + CLS, learned [MISSING] token. Cross-attention: gesture/motion queries attending to emotion/context keys. The rule-based T05 baseline = naive single-cue rules (wave→greet, angry→alert, thumbs up→success); the legend rubric is the compound mapping the fusion model must learn from data.
- Rows 19, 21, 48 are optional trims if the recording budget forces exactly 60 scenarios; quotas remain ≥ 1 train per intent per environment without them.

2. Legend

2.1 Intent classes (understandings, not robot actions)

Code	Name	Definition / boundary
F01	Greeting / positive acknowledgment	Person greets robot or signals genuine/playful positive acknowledgment (incl. task success).
F02	Emergency / danger	Hazard state: fire, burn, injury risk, panic, freeze response. Highest priority.
F03	Task assistance request	Person summons or directs robot to a task/object (beckon, point-and-move). Neutral/positive affect.
F04	Help request	Person in difficulty actively signals need (raise hand, beckon, thumbs down at task) with negative/neutral affect.
F05	Engaged / busy — no interaction needed	Person focused on own task or addressing someone else; robot must not interrupt.
F06	Requests passage / space	Person needs robot to move aside (approach on path, dismissal wave, push-aside).
F07	Frustration / agitation	Anger or agitation without physical danger; needs calm de-escalation, not alarm.
F08	Break / relief request	Aversion or fatigue toward the task; person wants relief or alternative (disgust-driven).
F09	Farewell	Person leaving the interaction/session (wave + motion toward exit).
F10	Discouraged / giving up	Sad disengagement without any directed signal (gesture = none, low energy).

2.2 Robot actions (unified across both environments)

Code	Action
A01	Positive acknowledgment; prompt/prepare next task
A02	Halt all motion + fire/smoke alert
A03	Halt all motion + medical alert
A04	Approach/follow to indicated spot; await instruction
A05	Offer task guidance / answer, supportive tone
A06	Hold position; do not interrupt; stay aware
A07	Acknowledge; suggest break/alternative activity
A08	Calm assistance, soft tone (de-escalation)
A09	Wave back; do not follow
A10	Acknowledge farewell; hold position
A11	Move aside promptly; clear path
A12	Encourage; suggest alternative approach
A13	Follow; offer to fetch/carry
A14	Halt risky action; check surroundings; notify supervisor
A15	Ask clarifying question

Multi-action rows (A02/A03): the intent classifier outputs F02; the policy layer selects A02 vs A03 (vs A14 in the classroom) from the hazard type visible in scene context. F09 allows A09 or A10 depending on whether the robot was engaged with the person.

2.3 Research goals

Code	Goal
G1	Fusion necessity: all four cues must be weighed jointly; any single-cue rule misfires.
G2	Robustness: correct intent under missing/degraded cues (sensor loss, occlusion, blur).
G3	Generalization: context flips, emotion–gesture conflicts, unseen cue combinations.

2.4 Test types

Code	Test
T01	Full-cue fusion accuracy (all modalities present, joint reasoning required)
T02	Cue-conflict resolution (contradictory emotion vs gesture)
T03	Missing-cue robustness / graceful degradation
T04	Context generalization (meaning flips with, or is invariant to, environment)
T05	Rule-based baseline comparison (naive per-cue rules fail; fusion model must not)

2.5 Cue vocabularies (fixed by the perception models)

Channel	Values
Emotion (7)	neutral, happy, sad, angry, disgust, fear, surprise
Gesture (7 + none)	raise hand, wave, point, beckoning, thumbs up, thumbs down, both hands up, neutral (none)
Motion (6)	sit, stand, walk, run, step back, lean forward

Channel	Values
Context (2)	Classroom, Kitchen (training) / Hospital, Museum, Shop (other environments)

Notation: '—' = not applicable. [MISSING] = the cue was NOT captured (sensor loss/occlusion) — listed in the Missing column. 'none' (gesture) = the idle class, a real captured observation. The fusion model must treat none and [MISSING] differently.

2.6 Labeling rubric (class boundaries — also the spec for the compound mapping the fusion model must learn)

- beckoning: neutral → F03; negative affect (sad) → F04
- point: task-embedded/static at own work → F05; at object → F03; at robot while approaching → F06; at object angrily, stationary → F07
- wave: toward robot → F01; toward exit → F09; angry (dismissal) → F06
- thumbs up: happy → F01; angry → F07 (sarcasm); disgust → F08 (sarcasm)
- thumbs down: sad + static → F04; angry → F07; disgust → F08; happy → F01 (playful)
- none (no gesture): sad + low energy → F10; neutral + task motion → F05
- both hands up: fear/surprise → F02; angry + stationary → F07; neutral + seated → F05 (stretch)
- raise hand: classroom + neutral → F04; classroom + happy → F05 (teacher-directed); kitchen + happy → F01
- motion/direction: fear/surprise + run/step back → F02; walk toward robot (hands occupied or angry) → F06; walk toward exit + wave → F09
- missing cues: rely on remaining cues + direction; if ≥ 2 cues missing and remainder is non-informative → default F05 (safe); if direction still indicates approach/hazard → F06/F02

3. Scenario table (62 rows)

Shading: amber = NEW scenario to record (25). Blue = kept from V2 with a correction (relabel, move, gesture/action/justification fix). White = unchanged from V2.

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
1	Train	Classroom	Student waves at robot with happy face while walking toward it.	happy	wave	walk	—	F01	G1, G3	Wave + happy alone is ambiguous (greet vs farewell). Motion direction toward robot is the disambiguator; the naive rule wave→greet only works by luck here and fails on #30.	A01	—
2	Train	Classroom	Student gives thumbs up with happy face while sitting.	happy	thumbs up	sit	—	F01	G1, G3	Emotion and gesture align → positive acknowledgment. Alignment anchor for the conflict rows (#33 train; #28/#29/#60 test): emotion resolves gesture polarity.	A01	—
3	Train	Classroom	Student suddenly stands up from chair, startled, throwing both hands up.	surprise	both hands up	stand	—	F02	G1	Sudden stand + both hands up + surprise = alarm state; robot halts and checks. Contrast #14: the angry variant is frustration (F07) — emotion changes the class.	A14	—
4	Train	Classroom	Student jumps back from a desk frightened, both hands up.	fear	both hands up	step back	—	F02	G1	Adds fear to the classroom (absent in V2). Fear + recoil from object + hands up = hazard response; classroom context routes the alert to the teacher (A14), not a fire dispatch.	A14	—
5	Train	Classroom	Student beckons robot over with neutral face while sitting.	neutral	beckoning	sit	—	F03	G1, G3	Rule neutral→no action misses an explicit summons. Beckoning must not be suppressed by neutral emotion; all cues are weighed jointly.	A04	—

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
6	Train	Classroom	Student beckons the robot to their desk with neutral face, seated — room-identity sensor offline.	neutral	beckoning	sit	context	F03	G2	Context masked in training: beckon + neutral + seated is still a summons (beckoning is context-free). Trains context-dropout robustness; test #24 uses a different tuple.	A04	—
7	Train	Classroom	Student raises hand with neutral face while sitting at desk.	neutral	raise hand	sit	—	F04	G1	All four cues needed: raise hand + sit + classroom + neutral = academic help request. The same gesture flips class by emotion (#11) and by context (#53).	A05	—
8	Train	Classroom	Student with sad face gives thumbs down toward own worksheet, barely moving, seated.	sad	thumbs down	sit	—	F04	G1	Sad + thumbs down at own work + static = distress signal. Minimal pair with #15 (angry variant → F07): emotion disambiguates identical gesture+motion.	A05	—
9	Train	Classroom	Student slumped and sad, staring down — hands below desk, no gesture captured.	sad	[MISSING]	sit	gesture	F04	G2	Gesture occluded (hands below camera). Sad + slumped + classroom still → help request. Moved from V2 Test so the missing-gesture pattern is learnable; tests now use #23/#54.	A05	—
10	Train	Classroom	Student writes notes during lecture, occasionally pointing at the notebook, neutral face.	neutral	point	sit	—	F05	G1, G3	Rule neutral+sit→idle/interruptible. Wrong: task-directed point at own notes = focused work; robot must suppress engagement.	A06	—
11	Train	Classroom	Student smiles and raises hand while sitting — volunteering to answer the teacher.	happy	raise hand	sit	—	F05	G3	Interaction is teacher-directed, not robot-directed. Rule raise-hand→F04 misfires; happy + raise hand + classroom = participation. (Justification added in V3; was empty in V2.)	A06	—
12	Train	Classroom	Student walks toward robot in a narrow aisle carrying books — hands occupied, gesture not captured.	neutral	[MISSING]	walk	gesture	F06	G2	Hands occupied → gesture channel lost. Walk toward robot in a narrow aisle + neutral = passage request; trains attention shift to motion + direction. (Justification added in V3.)	A11	—
13	Train	Classroom	Student, annoyed by the robot hovering, waves it away with angry face, standing.	angry	wave	stand	—	F06	G1, G3	Training conflict combo: rule wave→greet fires the opposite response. Angry + wave = dismissal → clear space. Test conflicts use different combos (#28/#29/#60/#62).	A11	—
14	Train	Classroom	Student raises both hands in frustration, rigid posture, standing.	angry	both hands up	stand	—	F07	G1, G3	Standing (not fleeing) + classroom = controlled frustration, not danger; de-escalate, don't alarm. RELABELED from F02 in V2 — the V2 justification already described de-escalation and the cues match kitchen twin #45.	A08	—
15	Train	Classroom	Student crumples a worksheet and gives thumbs down with angry face, seated.	angry	thumbs down	sit	—	F07	G1	Minimal pair with #8: identical gesture+motion, emotion flips intent (sad→help F04, angry→frustration F07). Gesture-only rules cannot separate them.	A08	—
16	Train	Classroom	Student pushes worksheet away, gives thumbs down with disgusted face, stepping back from desk.	disgust	thumbs down	step back	—	F08	G1, G3	Rule disgust→check smell/food is context-blind. Disgust + push-away + step back in a classroom = task aversion → break request.	A07	—
17	Train	Classroom	Student grimaces at a long assignment, gives thumbs down, slumping back in chair.	disgust	thumbs down	sit	—	F08	G1	Disgust (aversion) vs sad (distress, #8) under the same gesture+motion carries the F08/F04 boundary. Emotion granularity, not gesture, decides.	A07	—
18	Train	Classroom	Student waves goodbye with happy face while walking to the door at dismissal.	happy	wave	walk	—	F09	G1, G3	Minimal pair with #1: same emotion+gesture, direction flips intent (toward robot = greet, toward exit = farewell). Direction is the learned disambiguator.	A09	—

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
19	Train	Classroom	Student packs bag and waves briefly with neutral face, heading toward the door.	neutral	wave	walk	—	F09	G1	Neutral farewell variant: leave-taking must not require positive affect. (Optional trim if the recording budget is tight.)	A10	—
20	Train	Classroom	Student pushes assignment aside and rests head on desk, sad, no gesture, motionless.	sad	none	sit	—	F10	G1	'none' is a captured value, not a missing one: confirmed no gesture + sad + head-down = disengagement. vs #8 (thumbs down = active signal → F04). Gesture absence separates giving-up from help-seeking.	A12	—
21	Train	Classroom	Student stands by desk staring at a failed test, sad, arms hanging, no gesture.	sad	none	stand	—	F10	G1	Standing variant of #20; with #51/#52 this gives gesture-none coverage across contexts and postures. (Optional trim if budget-tight.)	A12	—
22	Test	Classroom	Student waves while walking toward robot — face occluded by a held book, emotion not captured.	[MISSING]	wave	walk	emotion	F01	G2	Emotion masked: wave + direction (toward robot) must yield greeting; the same masked pattern with exit direction is farewell (#30). Rules tie-break wrong without emotion.	A01	T03, T05
23	Test	Classroom	Student runs toward the exit in panic — gesture lost to motion blur.	fear	[MISSING]	run	gesture	F02	G2	Safety-critical degradation: fear + run + classroom must escalate without gesture. Mirrors kitchen #54 in the other environment.	A14	T03, T05
24	Test	Classroom	Student beckons robot while walking toward the supply shelf — room-identity sensor offline.	neutral	beckoning	walk	context	F03	G2	Context masked at test: beckon + walk toward object is still a task summons. Train counterpart #6 uses a different motion — the pattern, not the tuple, must transfer.	A13	T03, T05
25	Test	Classroom	Student raises hand leaning forward — emotion sensor occluded.	[MISSING]	raise hand	lean forward	emotion	F04	G2	Emotion masked; gesture + posture + context suffice. Tests attention redistribution to the remaining cues; robot clarifies rather than assumes (A15).	A15	T03, T05
26	Test	Classroom	Student stretches both arms overhead mid-study with neutral face, then returns to writing, seated.	neutral	both hands up	sit	—	F05	G3	Hard negative for F02: every training both-hands-up row is alarmed or standing (#3,#4,#14,#34,#35,#45). Neutral + seated + classroom = a stretch. Rule both-hands-up→emergency misfires.	A06	T01, T05
27	Test	Classroom	Student stands up angrily, points at robot, stepping toward it.	angry	point	walk	—	F06	G1, G3	Rule angry+point→threat alert. Approach + point at robot = space/passage demand; direction + gesture target resolve it. De-escalating tone retained in the action.	A11	T01, T04, T05
28	Test	Classroom	Student scowls angrily while giving an exaggerated thumbs up after being corrected, seated.	angry	thumbs up	sit	—	F07	G3	Conflict-transfer test: angry overrides thumbs up (sarcastic compliance). Training conflicts are different combos (#13/#33); classroom seated variant of #60 checks context invariance.	A08	T02, T04, T05
29	Test	Classroom	Student gives thumbs up with disgusted face after finishing a disliked task, seated.	disgust	thumbs up	sit	—	F08	G3	Sarcastic completion: disgust overrides thumbs up → relief, not praise. Strongest naive-rule failure (thumbs up→success). Action filled in V3 (was empty).	A07	T02, T05
30	Test	Classroom	Student waves with neutral-looking face while walking out of the classroom — emotion not detected.	[MISSING]	wave	walk	emotion	F09	G2, G3	Emotion masked: wave + exit direction → farewell; rule fires greet. Train counterpart #50 (kitchen) — same pattern, different environment.	A09	T03, T05
31	Test	Classroom	Student drops the crumpled quiz on the desk and stands turned away, sad, arms slack, no gesture.	sad	none	stand	—	F10	G1	Discouragement with mild variation (turned away, standing): no directed gesture, sad, low energy.	A12	T01, T05

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
										Must not divert to F04 (no signal present) or F09 (no wave).		
32	Train	Kitchen	Cook tastes finished dish, smiles, gives thumbs up, standing still.	happy	thumbs up	stand	—	F01	G1	All cues align: genuine completion → positive acknowledgment; alignment anchor for the kitchen conflict rows (#33 train, #60/#61 test).	A01	—
33	Train	Kitchen	Person gives thumbs down while smiling and shaking head playfully — family tasting game.	happy	thumbs down	stand	—	F01	G3	Happy overrides thumbs down → playful banter, not distress. MOVED from V2 Test so training contains one conflict example; action corrected A08→A01 (V2 assigned the de-escalation action to an F01 row).	A01	—
34	Train	Kitchen	Person extends both hands out to signal stop and runs back from smoke at the stove.	fear	both hands up	run	—	F02	G1	Maximal emergency: fear + stop-hands + fleeing smoke. Fusion adds redundancy for noisy cues. Motion normalized to run + direction away from object (V2: 'Move backward (run)').	A02/A03	—
35	Train	Kitchen	Person touches a hot pan, startled, throws both hands up and steps back.	surprise	both hands up	step back	—	F02	G1	Surprise + recoil at hot pan = burn risk. The same cue pattern in the classroom routes to A14 (no fire/medical dispatch) — context selects hazard handling.	A02/A03	—
36	Train	Kitchen	Person beckons robot toward the stove with neutral face while walking there.	neutral	beckoning	walk	—	F03	G1, G3	Beckon + walk toward stove = assistance at destination; neutral emotion must not suppress. Direction toward object separates this from F06 (toward robot).	A04	—
37	Train	Kitchen	Person points at a shelf while walking toward it, neutral face.	neutral	point	walk	—	F03	G1, G3	Point + walk toward shelf = fetch/carry request. The direction field distinguishes this from #27 (point + toward robot = give way).	A13	—
38	Train	Kitchen	Person slumped against counter, sad, barely moving, gives a weak thumbs down.	sad	thumbs down	sit	—	F04	G1	Static distress at the counter → guidance. Sad + directed signal (thumbs down) + static = F04 per rubric; contrast #51 (no gesture at all → F10).	A05	—
39	Train	Kitchen	Person beckons robot with tired, sad face while standing over a pile of dirty dishes.	sad	beckoning	stand	—	F04	G1	Beckon + negative affect = explicit help request. Train counterpart of test #57 (same combo with context masked) — makes that inference learnable rather than zero-shot.	A05	—
40	Train	Kitchen	Person chops vegetables with neutral focus, back to robot, occasionally pointing at ingredients.	neutral	none	stand	—	F05	G1, G3	Rule neutral+stand→available. Task-embedded point (ingredients) + kitchen = focused work; interrupting near knives is unsafe.	A06	—
41	Train	Kitchen	Person wipes hands on apron, neutral, glancing around — transition between tasks, self-directed point.	neutral	none	stand	—	F05	G1, G3	Self-directed (inward) gesture during a task transition ≠ invitation. Do not engage.	A06	—
42	Train	Kitchen	Person pushes robot aside with disgusted face, pointing it away, stepping back.	disgust	point	step back	—	F06	G1, G3	Robot-directed point + stepping back = clear-path demand; rule disgust+step-back→hazard-check misfires. Cross-environment pair with #27.	A11	—
43	Train	Kitchen	Person carrying a hot pot with both hands walks toward robot in the galley aisle — no gesture possible.	neutral	[MISSING]	walk	gesture	F06	G2	Hands occupied by a hot pot → gesture missing; walk toward robot in the aisle = urgent passage. Kitchen counterpart of #12 — the missing-gesture pattern exists in both contexts.	A11	—
44	Train	Kitchen	Person slams hand on counter then raises both hands, angry, rigid, standing.	angry	both hands up	stand	—	F07	G1, G3	Slam then both hands up (scenario text now includes the recorded gesture): stationary + kitchen = venting, not danger. Rule angry→emergency over-escalates.	A08	—

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
45	Train	Kitchen	Person angrily points repeatedly at spilled food on the floor while looking at the robot, standing.	angry	point	stand	—	F07	G1, G3	The point targets the mess, not the robot's space; stationary = complaint, not aggression. Minimal pair with #27: walking-toward flips it to F06.	A08	—
46	Train	Kitchen	Person recoils from refrigerator odor, disgusted, thumbs down, stepping back.	disgust	thumbs down	step back	—	F08	G1, G3	Rule disgust+kitchen→inspect food acts on the object; the person-centric reading (aversion + withdrawal → relief) is correct.	A07	—
47	Train	Kitchen	Person stands at the sink grimacing at greasy pans, gives thumbs down, leaning away.	disgust	thumbs down	stand	—	F08	G1	Standing variation of #47 for recording diversity; keeps the rubric airtight (disgust→F08). (Optional trim if budget-tight.)	A07	—
48	Train	Kitchen	Person waves at robot with happy face while walking toward the exit.	happy	wave	walk	—	F09	G1, G3	Wave + happy is ambiguous with greeting; exit direction resolves it. Single action A09 (V2 listed A09+A10; A10 remains the alternative in the legend).	A09	—
49	Train	Kitchen	Person unties apron and waves while walking to the door, face turned away — emotion not captured.	[MISSING]	wave	walk	emotion	F09	G2	Emotion-missing farewell in training (face turned away): wave + direction suffice. Train counterpart of test #30.	A10	—
50	Train	Kitchen	Person sits down heavily, sad, staring at a failed dish, hands in lap — no gesture.	sad	none	sit	—	F10	G1	Gesture corrected thumbs down→none: V2's own justification said 'no reaching gesture', and the V2 tuple collided with #38 (identical cues → F04). Sad + no gesture + sitting = giving up. VERIFY against recordings; if actors performed thumbs down, re-record without it.	A12	—
51	Train	Kitchen	Person stares at a burnt cake, shoulders dropped, sad, hands at sides, standing motionless.	sad	none	stand	—	F10	G1	Standing variant; completes gesture-none coverage in the kitchen. Contrast #38: a directed signal → F04.	A12	—
52	Test	Kitchen	Person raises hand and smiles while sitting at the kitchen table after eating.	happy	raise hand	sit	—	F01	G3	Context flip: the same cues are participation/help in the classroom (#7 neutral, #11 happy) but social acknowledgment in the kitchen. Tests context-conditioned meaning.	A01	T04, T05
53	Test	Kitchen	Person runs from the stove area in fear — gesture lost to motion blur.	fear	[MISSING]	run	gesture	F02	G2	Gesture lost to blur; fear + run + kitchen must still escalate. Safety-critical robustness; train counterparts #12/#44 plus modality dropout.	A02/A03	T03, T05
54	Test	Kitchen	Person stands frozen, wide-eyed, staring at the oven — rigid, no gesture.	fear	none	stand	—	F02	G3	Freeze response: all training emergencies are dynamic (run/recoil); fear + rigid stillness + oven must still escalate. V2 had emotion missing AND stare-as-'point' — unanswerable from captured cues; V3 keeps emotion observable so the row is resolvable.	A02/A03	T01, T05
55	Test	Kitchen	Person cheerfully points at a high cupboard while walking toward it, looking at robot.	happy	point	walk	—	F03	G3	Unseen emotion for F03 (happy): task requests must be emotion-invariant. Walk toward object + point = fetch; distinct from #40/#42 (static, task-embedded point → F05).	A13	T01, T05
56	Test	Kitchen	Person beckons robot with sad face, standing — room-identity sensor offline.	sad	beckoning	stand	context	F04	G2	Context masked: sad + beckon suffice for a help request. Train counterpart #39 (context present) makes the inference learnable, not zero-shot.	A05	T03, T05
57	Test	Kitchen	Person stands neutral, no gesture visible, stationary — emotion sensor noisy.	[MISSING]	[MISSING]	stand	emotion, gesture	F05	G2	Worst-case double masking: only motion + context remain. Correct behavior is the safe default (F05), not a misfired active intent.	A06	T03, T05

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Int.	Goal	Why naive rules fail / cue justification	Action	Test
58	Test	Kitchen	Person carrying stacked trays walks toward robot — hands occupied and face hidden behind the trays.	[MISSING]	[MISSING]	walk	emotion, gesture	F06	G2	Double masking with an ACTIVE ground truth: walk toward robot in the aisle must yield give-way, not the F05 fallback. Separates true robustness from blanket defaulting; direction is the deciding cue.	A11	T03, T05
59	Test	Kitchen	Person gives thumbs up with angry face after being told to redo a recipe, standing.	angry	thumbs up	stand	—	F07	G3	Sarcastic compliance: angry overrides thumbs up. Different conflict combo from training (#13/#33) — tests interaction generalization, not memorization.	A08	T02, T05
60	Test	Kitchen	Person pushes plate away at the table, disgusted, thumbs down, seated.	disgust	thumbs down	sit	—	F08	G1, G3	Seated aversion variant; cross-context pair with #16 (classroom, step-back). Rule disgust-in-kitchen→inspect-food misfires.	A07	T01, T04, T05
61	Test	Kitchen	Person waves while walking toward the exit with sad face after a failed cooking session.	sad	wave	walk	—	F09	G3	Unseen combo sad+wave: farewell must be driven by gesture + direction, emotion-invariant; must not divert to F04 despite sadness. Naive happy-wave rules fail.	A10	T02, T05
62	Test	Kitchen	Person slumps forward over the counter, head down over the failed dish, sad, no gesture.	sad	none	lean forward	—	F10	G3	Unseen posture (lean forward) for the learned F10 pattern (sad + no gesture + low energy, #20/#21/#51/#52). Tests posture and context generalization.	A12	T01, T04, T05

4. Change log (V2 → V3)

- V2 #5 → V3 #14: relabeled F02→F07 (V2 justification already described de-escalation; matches kitchen twin V2 #24).
- V2 #14 → V3 #9: moved Test→Train (the missing-gesture pattern must be learnable; #23/#54 keep it tested).
- V2 #39 → V3 #33: moved Test→Train (training needs at least one emotion–gesture conflict); action corrected A08→A01.
- V2 #28 → V3 #51: gesture thumbs down→none — V2's own justification said 'no reaching gesture', and the V2 tuple was identical to V2 #21 with a different label (irreducible label collision). VERIFY the recordings; if thumbs down was performed, re-record without it.
- V2 #38 → V3 #55: 'point (frozen stare direction)'→none (a stare is not a gesture) and emotion [MISSING]→fear — with both gone the row was unanswerable from captured cues.
- V2 #11 → V3 #11: empty justification filled (teacher-directed participation); action 'Give a chance'→A06.
- V2 #8 → V3 #12: empty justification filled.
- V2 #13/#15/#17 → V3 #25/#29/#30: empty robot actions filled (A15/A07/A09).
- V2 #19 → V3 #34: motion 'Move backward (run)' normalized to run + direction 'away from object'; V2's dual action A02+A03 kept as 'A02/A03' with the legend rule: hazard type from scene context selects between them.
- V2 #25 → V3 #49: single action A09 (V2 listed A09+A10).
- V2 #3,#4,#6,#9,#21,#22,#24,#37 → scenario texts updated to explicitly mention the recorded gesture (recording practice: actors perform the listed gesture).
- V2 skipped row #10; V3 renumbered 1–62 grouped by environment and split.
- Intent names unified (F02 and F08 had two names each); all cue values lower-cased; Direction column added (motion direction was previously buried in scenario prose despite being the stated disambiguator).
- 25 NEW scenarios added (shaded amber) to reach full intent×environment×split coverage: every intent now has ≥2 train + 1 test per environment (F02/F04/F05 carry extras).
- Rows 19, 21, 48 marked optional if the recording budget forces exactly 60 scenarios.

5. Automated verification results

- Totals: 62 scenarios — 41 train / 21 test (2:1); Classroom 31 (21/10), Kitchen 31 (20/11). All 10 intents appear in both environments with ≥ 2 train + ≥ 1 test each; per intent overall: ≥ 4 train, 2 test (F02: 3 test — safety-critical class intentionally over-tested).
- Vocabulary: every emotion/gesture/motion/context value in the table belongs to the fixed cue vocabularies; Missing column always agrees with [MISSING] cell markers; every action code is defined in the legend.
- No label collisions: no two rows share an identical observable cue tuple with different intents (the V2 #21 vs #28 collision was repaired).
- No train–test leakage: no identical cue tuple appears in both splits; every conflict combo in test (angry+thumbs up, disgust+thumbs up, sad+wave) is absent from training, whose conflicts are happy+thumbs down and angry+wave — conflict resolution is tested as generalization, not recall.
- Missing-cue coverage: train contains each single-cue missing pattern that appears in test (emotion, gesture, context). The double-missing test rows (#58, #59) are covered by modality-dropout augmentation; simulation confirms #58 is genuinely ambiguous under double masking, so its F05 safe-default label is the only defensible one, while #59 stays identifiable through direction (toward robot).
- Rubric lint: all 62 rows are consistent with the labeling rubric in section 2.6 — zero contradictory labels.
- Coverage: gesture 'none' appears in training in both environments (#20, #21, #51, #52); fear appears in both environments; every direction value occurs in training.
- Action item before recording: verify V3 #51 (V2 #28) against existing videos — if actors performed thumbs down, re-record without it, since sad+thumbs down+sit is reserved for F04 (#8, #38).